Cognitive Behavior in Cephalopod-inspired Biomimetic Robots

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Introduction

Project Goals

- Build the OctoBot, a bio-inspired soft robot that classifies objects by touch using a single embedded pressure sensor
- Investigate how an autonomous agent learns to choose between immediate and delayed rewards

Background

- Octopuses identify objects through receptors in their tentacles, which remains difficult to replicate this in low-cost soft robot
- Cuttlefish have memory-guided delayed gratification behavior that can inspire autonomous robot learning

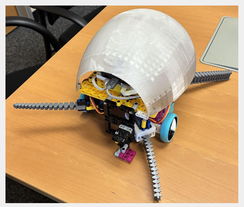
Contributions:

- Soft tentacle with integrated flexible pressure sensing
- Random Forest classifier achieving 91% grouped accuracy
- Full pipeline deployed on ESP32 with real-time LED feedback
- Combined reinforcement learning, computer vision, and reward-based navigation into a decision-making task on a biologically inspired robotic framework

Materials

OctoBot

- **Robot Platform:** LEGO SPIKE Prime base, LMS-ESP32 controller
- **Tactile Sensing:** flexible resistive pressure strip in a 95A TPU tentacle actuated by servo motor winch
- **Object Set:** nine objects varying in size, density, and texture



Delayed Gratification

- **Robotic Base and Control:** The delayed gratification agent is a TurtleBot4 Lite, which uses a Raspberry Pi 4B for central processing
- **Visual Processing:** Built-in camera on the TurtleBot4
- **Software Stack:** ROS2, Ubuntu for robot communication and SLAM for mapping and navigation

Environment:

- **Seagrass:** Construction paper of various shades of green are folded and cut into natural, grass like forms
- **Delayed gratification chambers:** The chambers are constructed out of wood, with clamp holder clips attached and an acrylic board used to create a clear sliding door
- **Rocks:** Created out of stacked foam boards, sculpted to create realistic texture and form
- **Coral:** Made of a wire frame wrapped with masking tape and coated with paper mache



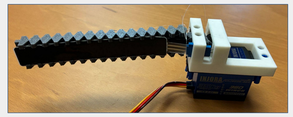
Methods

Octobot

Data Collection: servo winch pulls on fishing line curling the tentacle and producing a pressure time-series (25 grips per object)

Feature extraction: 27 features covering compression depth, temporal dynamics, and spectral content

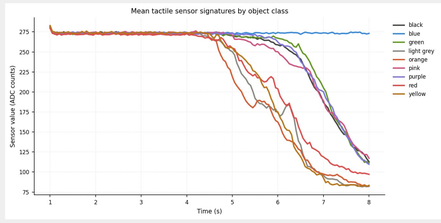
Classification: Random Forest (200 trees), q0-fold cross-validation, evaluated on 9-class and grouped 3-class tasks



Tentacle Setup



Object Set



Mean Sensor Signals by Object

Delayed Gratification

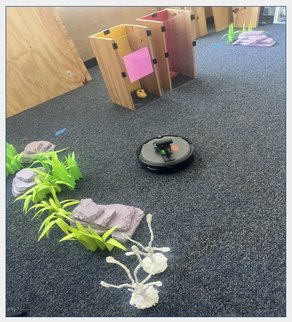
Experimental Design: A Q-learning agent was trained in a two-choice delayed gratification task inspired by cuttlefish foraging. Reward locations switched across trials, so the agent had to learn reward value rather than a fixed side preference. After training, the agent was tested across delays from 10 to 130 seconds, and the learned Q-table was also transferred to a TurtleBot for navigation-based reward choice.

State	Condition	Left option	Right option	Expected best action
State 0	Experimental	Live shrimp	Dead shrimp	Left
State 1	Experimental	Dead shrimp	Live shrimp	Right
State 2	Control	Unobtainable	Dead shrimp	Right
State 3	Control	Dead shrimp	Unobtainable	Left

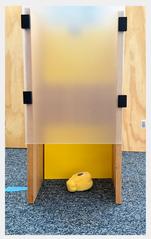
Table 1. Task states used in the delayed gratification experiment.

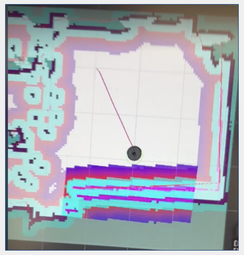
Robot Implementation:

- The robot was placed in a two-choice arena containing a left and right chamber
- A SLAM-generated map of the arena was used for localization and autonomous navigation, allowing the robot to move between a defined home position, a central decision point, and the left or right reward chambers.



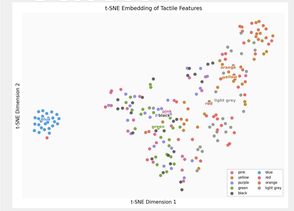




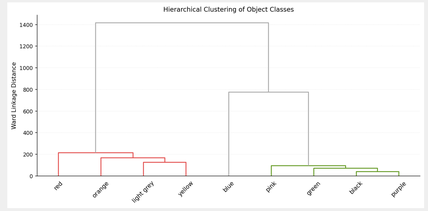


Results and Discussion

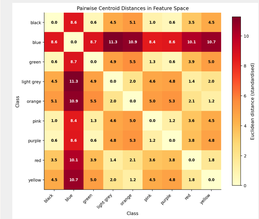
Octobot



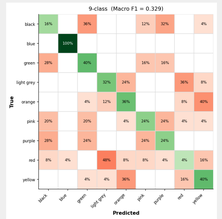
t-SNE Plot Indicating Groupings



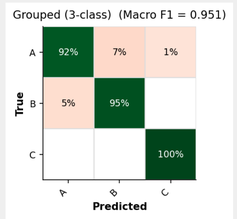
Hierarchical Clustering of Objects



Heatmap of Centroid Distances



Individual Random Forest



Grouped Random Forest

Tactile pressure signals alone are sufficient to reliably distinguish broad material categories, supporting feasibility of touch-based object identification in soft robotic systems.

Delayed Gratification

In **Figure 1**, the agent learned the correct action for each task state over training trials. Experimental states show learning of the preferred live shrimp option, while control states show learning to avoid the unobtainable option. In **Figure 2**, the agent preferred the delayed high-value reward at short delays, but this preference decreased as delay increased. Both figures represent results from simulated trials.

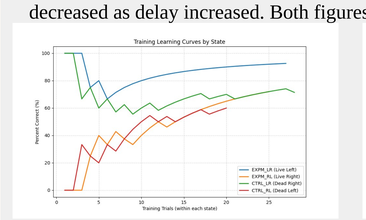


Figure 1. Training Learning Curves by State

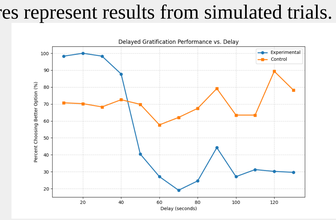


Figure 2. Delayed Gratification Performance vs. Delay

Conclusion: Overall, this model demonstrates that a Q-learning agent can learn flexible, value-based delayed gratification behavior in simulation and transfer the learned choice policy to a TurtleBot for embodied navigation-based decision making.

Simulation Q-table

State	Q(LEFT)	Q(RIGHT)
State 0	4.71	0.27
State 1	0.57	3.73
State 2	0.40	0.65
State 3	0.75	0.34

Robot Training Q-table

State	Q(LEFT)	Q(RIGHT)
State 0	3.588	0.469
State 1	0.410	4.074
State 2	0.433	0.771
State 3	0.718	0.235

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